MILITARY SPECIFICATION

DARKROOM, PHOTOGRAPHIC, TRANSPORTABLE ES-82( )

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agenices of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of truck transportable photographic darkroom designated Darkroom, Photographic, Transportable ES-82( ) (see 6.6).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-S-86	- Safelight, Darkroom, Photographic.
RR-T-646	- Trays, Processing, Photographic.

MILITARY

MIL-P-116	- Preservation, Methods Of.
MIL-P-11268	- Parts, Material, And Processes Used In Electronic Equipment.
MIL-H-11511	- Heater, Space, Multifuel, With Blower, 60,000 Btu/hr.
MIL-M-13231	- Marking Of Electronic Items.
MIL-F-14072	- Finish For Ground Electronic Equipment.
MIL-A-14372	- Air Conditioner, Base-Mounted, Self-Contained Type, Air-cooled, Nominal, 9,000 Btu/hr.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703-5304, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6780

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

MIL-S-55286	- Shelter, Electrical Equipment S-280( )/G.
MIL-S-55507	- Shelter, Electrical Equipment (With Or Without Equipment), Packaging Of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-252	- Classification Of Visual And Mechanical Defects For Equipment, Electronic, Wired, And Other Devices.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission And Susceptibility Requirements For the Control of Electro-Magnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-810	- Environmental Test Methods.
MS27130	- Nut, Plain, Blind Rivet-Flat and Countersunk Head, Open End.

2.1.2 Other government documents. The following government drawings form a part of this specification to the extent specified herein:

DRAWINGS

ELECTRONICS COMMAND

DL-SM-B-682112	- Darkroom, Photographic, Transportable ES-82( ).
DL-SM-B-682701	- Temperature Control Unit, Photographic Processing FH-19( ).
DL-SM-B-682798	- Heater, Water.
DL-SM-B-683824	- Air Conditioner Duct.
DL-SM-B-682825	- Heater Dryer Section.
DL-SM-B-682826	- Air Conditioner Installation.
DL-SM-B-682827	- Escape Door Installation.
DL-SM-B-682828	- Plumbing Installation.
DL-SM-B-682829	- Shelf Installation Forward Compartment.
DL-SM-B-682853	- Partition Installation.
DL-SM-B-682854	- Installation-Counters, Shelves And Angles, Center Compartment.
DL-SM-B-682856	- Installation Shelf Cabinet Refrigerator.
DL-SM-B-682857	- Lighting Fixture.
DL-SM-B-682858	- Installation Of Typewriter.
DL-SM-B-682859	- Cabinet Assembly.
DL-SM-B-682860	- Installation Of Enlarger EN-105( ) And Processor EH-91( ).

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currentversion before use.

DL-SM-B-682861 - Installation Of Storage Box.
DL-SM-B-682881 - Printer, Projection, Photographic EN-105( ).
DL-SM-B-682882 - Processing Machine, Photographic Paper EH-91( ).
DL-SM-B-682902 - Processing Unit, Photographic Film EJ-12( ).
DL-SM-B-682903 - Processing Unit, Photographic Film EJ-13( ).
DL-SM-B-682908 - Intercommunication Station LS-586/U.
DL-SM-B-682909 - Safelight, Darkroom, Photographic FM-188( ).
DL-SM-B-682911 - Timer, Interval FN-5( ).
DL-SM-B-682912 - Timer, Interval FM-17( ).
SC-D-82768 - Aircraft Loading Data Plates For Trailer, Semi-Trailer, Shelter Truck and Equipment.
SC-DL-108736 - Ladder, Vehicle Boarding MX-3391( )/G.
SM-C-682129 - Cover, Air Conditioning.
SM-C-682134 - Cable, Power.
SM-C-682135 - Cable, Power.
SM-D-682140 - Storage Tank Assembly.
SM-E-682329 - Sink Installation.
SM-D-682338 - Decal.
SM-C-682436 - Dehumidifier Assembly Installation.
SM-C-682477 - Decal.
SM-D-682481 - Decal.
SM-D-682521 - Installation, Sink Stowage.
SM-C-682523 - Cover, Processor EN-91( ).
SM-D-682524 - Cover, EN-105( ).
SM-E-682593 - Installation-Stencil.
SM-D-682598 - Installation-Floor Covering.
SM-C-682850 - Decal.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70-28 - National Electrical Code - 1978.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS.

3.1 Description. The darkroom shall be a truck transportable, complete self-contained photographic laboratory capable of developing, printing and captioning both photographic film and print.

3.1.1 Major components. The darkroom shall consist of the following major components.

Item	Paragraph
Intercommunication station LS-586/U	3.5.1
Shelter, electrical equipment S-280( )/G	3.6.1
Heater, space	3.6.2.1
Air conditioner	3.6.2.2
Lighting fixtures	3.6.3.4
Safelight, darkroom, photographic FM-188( )	3.6.3.5
Hot water heater	3.6.4.5
Temperature control unit, photographic processing FH-19( )	3.6.4.6
Processing unit, photographic film EJ-12( )	3.6.5.1
Processing unit, photographic film EJ-13( )	3.6.5.2
Printer, Projection, Photographic EN-105( )	3.6.6.1
Processing Machine, Photographic Paper EH-91( )	3.6.6.2
Timer, Interval FN-5( )	3.6.8
Timer, Interval FM-17( )	3.6.9

3.2 First article. The quantity of samples specified in the contract shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the darkroom to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification.

3.4 Parts, materials and processes; general. Parts, materials, and processes shall be in accordance with drawings specified in 3.5. If not covered on drawings, they shall be in accordance with the requirements of MIL-P-11268.

3.4.1 Threaded inserts. No sheet metal or self-threading screws shall be used on either the inside or outside skins of the shelter. Threaded inserts shall be in accordance with MS27130 and shall be installed in the mounting members. The head of the threaded inserts shall not protrude more than 0.062 inch (1.57mm) from the shelter skin. The installed inserts for 0.3125 inch (7.94mm) bolts shall withstand an axial load of 2,000 pounds (907kg) followed by a torque load of not less than 40 lbf. in. (4.5Nm). Unless otherwise specified on the drawings, inserts shall be 0.3125-18UNC.

3.5 Construction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.5.1 Physical construction. Physical construction shall be in accordance with the drawings and documents referenced in 3.6.

3.6 Detail requirements.

3.6.1 Shelter, electrical equipment S-280( )/G. An electrical equipment shelter S-280( )/G, conforming to MIL-S-55286, shall house the required equipment for the darkroom.

3.6.1.1 Escape door. An escape door shall be installed in the shelter in accordance with DL-SM-B-682827.

3.6.1.2 Floor covering. The floor and 6 inches (152.4mm) up the 4 shelter walls, shall be covered in accordance with SM-D-682598. The floor covering shall have no voids between the covering and wall or floor. After installation of equipment, all mounting points shall be sealed with a chemical-resistant sealant.

3.6.1.3 Partitions. The shelter partitions shall be installed in accordance with DL-SM-B-682853. The intercompartment doors shall open and close easily by sliding. The doors shall not sag or stick against the frame, nor shall the bottom edges of the doors strick the floor after the darkroom has been subjected to the environmental conditions and tests specified herein.

3.6.1.4 Shelves. Shelves in the front section shall be installed in accordance with DL-SM-B-682829. Shelves, counters and angles in the center section shall be installed in accordance with DL-SM-B-682854, except that the elastic stop nuts used in retaining the movable shelves shall be replaced with anchor nuts. Brackets and supports for shelves and cabinets shall be attached to the mounting members.

3.6.1.5 Ladder, vehicle boarding MX-3391( )/G. A vehicle boarding ladder, in accordance with SC-DL-108736 shall be furnished.

3.6.2 Heating, ventilating and air-conditioning system. A heating, ventilating, and air-conditioning system, installed in accordance with the following, shall regulate the air circulation and provide controlled heating, ventilation, cooling and relative humidity.

3.6.2.1 Heater, space. A multi-fuel, thermostatically-controlled hot air space heater, conforming to MIL-H-11511, shall be installed in accordance with DL-SM-B-682825. The heater shall maintain an internal temperature of not less than 60°F (16°C) with an external temperature of -40°F (-40°C).

3.5.2.2 Air conditioners. Two air conditioners, conforming to MIL-A-14372, shall be installed in accordance with DL-SM-B-682826. Covers in accordance with SM-C-682129 shall be provided to enclose the air conditioners when not in use. The air conditioners shall maintain an internal temperature of not less than 90°F (32°C) with a temperature difference not greater than 10°F (6°C) between the floor and ceiling, with an external temperature of 120°F (49°C).

3.6.2.3 Dehumidifier, space. A dehumidifier, 50/60 Hertz (Hz), 120 volts alternating current (vac), single phase, shall be furnished in accordance with SM-C-682436. The dehumidifier power cable shall be routed and secured to prevent accidental damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

The connector shall be replaced with 90 degree connector. Collapsible handles shall be installed on the dehumidifier. The dehumidifier shall maintain the internal relative humidity at 60 percent with an external relative humidity of 90 percent at 100°F (38°C).

3.6.3 Electrical system. The electrical system shall conform to the National Electrical Code NFPA 70-78 and shall be furnished in accordance with DL-SM-B-682112. The darkroom shall operate from a 120/208 vac ±5 percent, 60 Hz ±5 percent, 3 phase, 4 wire power source.

3.6.3.1 Identification of wiring. All internal electrical wiring, except pre-wired, off-the-shelf commercial items, shall be color-coded in accordance with the National Electrical Code NFPA 70-78, Article 210-5.

3.6.3.2 Breakdown voltage. There shall be no flashover or breakdown when the equipment is tested as specified in 4.6.2.1. All adjacent metal parts not in the circuit shall be considered as ground.

3.6.3.3 Power cables. The power cables and adapter cable shall be constructed in accordance with SM-C-682134 and SM-C-682135 respectively. Strain relief shall be provided for the main power cable and there shall be no interference between the power cable and the outside intercom door.

3.6.3.3.1 Insulation resistance. There shall be no flashover or breakdown when the assembled cables are tested as specified in 4.6.2.2.

3.6.3.4 Lighting fixtures. The lighting fixtures, constructed in accordance with DL-SM-B-682857, shall be installed in the shelter.

3.6.3.5 Safelight FM-188( ). The safelight FM-188( ) shall conform to GG-S-86 and DL-SM-B-682909 and shall be installed in the shelter.

3.6.4 Plumbing system. The plumbing system shall be installed in the shelter in accordance with DL-SM-B-682828.

3.6.4.1 Piping. All pipe joints and connections shall withstand a pressure of 90 lbf/in^2 (psi) (621 kPa). Joints that require disassembly and reassembly shall not be equipped with solder or weld-type fittings.

3.6.4.2 Water supply pump. The water supply pump shall be capable of delivering 5 gal/min (19 liters/min) against a 100 foot (30m) head pressure. The automatic pressure-activated switch shall stop and start the pump at high and low level limits of the water storage tank. The pump shall be equipped with a cleanable and non-disposable type strainer. The pump shall be capable of pumping water from a source 15 feet (4.57m) below the level of the pump through the entire plumbing system.

3.6.4.3 Pressure regulator. The adjustable pressure regulator shall reduce commercial water pressure to 40 psi (275kPa).

3.6.4.4 Storage tank. The storage and expansion tank, in accordance with SM-D-682140, shall have a capacity of 50 gallons (189 liters) of water.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

3.6.4.5 Hot water heater. The hot water heater, in accordance with DL-SM-B-682798, shall operate from a 115 vac, 60 Hz, and 4.5 kilowatts power source. The heater shall raise the water temperature from 33°F (0.6°C) to 150°F (66°C) in not more than 2 hours.

3.6.4.5.1 Liquid level switch. The liquid level switch shall shut off the hot water heater if the water level drops below the top of the heating element.

3.6.4.5.2 Pressure relief valve. The pressure relief valve shall release the pressure between 75 and 85 psi (517 and 586kPa).

3.6.4.6 Temperature control unit, photographic processing FH-19( ). The temperature control unit shall be in accordance with DL-SM-B-682701, except that the retaining screws of the top cover shall be quick opening fasteners. The temperature control unit shall maintain a water temperature within a range of 68°F (20°C) to 90°F (32°C) in the temperature-controlled compartment of the sink (see 3.6.4.7). The temperature control unit shall lower the temperature of 26 gallons (98 liters) of water from 90°F (32°C) to 68°F (20°C) in not more than 2 hours in an ambient temperature of 100°F (38°C). In addition, the water heater of the temperature control unit shall maintain the pre-set water temperature whenever that temperature fails below the thermostat setting.

3.6.4.6.1 Valve controls. The control valve for the wash tank shall be marked and identified. Universal joints and short shafts shall be installed to locate the operating handles for D5, D10, and D11 above the level of the sink top and against the roadside wall of the front compartment.

3.6.4.6.2 Thermostat control. The thermostat control of the temperature control unit shall maintain the temperature of the circulating water within a range of 68°F (20°C) to 90°F (32°C) with a tolerance on any preset temperature within ±2°F (+1.1°C).

3.6.4.6.3 Water circulating pump. The water circulating pump of the temperature control unit shall deliver 60 gal/hr (227 liters/hr) against a 4 psi (28 kPa) head pressure. The pump shall maintain a water circulating rate of 3 gal/min (11 liters/min).

3.6.4.7 Sink. A 2- compartment sink shall be installed in accordance with SM-E-682329. The sink shall provide storage in accordance with SM-D-682521. The left-hand compartment of the sink shall be connected to the photographic processing temperature control unit FH-19( ). This compartment shall hold the components of the photographic film processing unit, EJ-13( ).

3.6.4.8 Drainage. With the shelter in a level position and the discharge end of a 50 foot (15m), 1 inch (25.4mm) inside diameter hose, positioned not more than 2 feet (0.61m) lower than the floor of the shelter, drainage by gravity of all water, chemical solutions, etc., shall be completed in not more than 15 minutes.

3.6.5 Film processing equipment. The following equipment shall be used for processing exposed film in still camera lengths:

Processing unit, photographic film EJ-12( ) (see 3.6.5.1).
Processing unit, photographic film EJ-13( ) (see 3.6.5.2).
Heater-dryer (see 3.6.6.4).

3.6.5.1 Processing unit, photographic film EJ-12( ). The photographic film processing unit EJ-12( ), shall be furnished in accordance with DL-SM-B-682902.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

3.6.5.2 Processing unit, photographic film EJ-13( ). The photographic film processing unit EJ-13( ), shall be furnished in accordance with DL-SM-B-682903.

3.6.6 Print processing equipment. The following equipments shall be used for printing and processing black and white prints:

Printer, projection, photographic EN-105( ) (see 3.6.6.1).
Processing machine, photographic paper EH-91( ) (see 3.6.6.2).
Heater-dryer (see 3.6.6.4).
Processing tray PH-158( ) (see 3.6.6.5).
Processing tray, PH-161( ) (see 3.6.6.6.).
Captioning equipment (see 3.6.7).

3.6.6.1 Printer, projection, photographic EN-105( ). Two each photographic projection printers EN-105( ) shall be modified in accordance with DL-SM-B-682881 and shall be installed in the shelter in accordance with DL-SM-B-682860.

3.6.6.2 Processing machine, photographic paper EH-91( ). Two each photographic paper processing machines EH-91( ), in accordance with DL-SM-B-682882 (except that the split-type lockwasher under the grounding screw shall be replaced with a star-type washer), shall be installed in the shelter in accordance with DL-SM-B-682860. The processing machine shall operate on a power source of 115 vac, 50/60 Hz.

3.6.6.3 Dust covers. Two dust covers in accordance with SM-D-682524 shall be furnished for the photographic projection printers EN-105( ) and two dust covers in accordance with SM-C-682523 shall be furnished for the photographic paper processing machines EH-91( ).

3.6.6.4 Heater-dryer. A heater-dryer cabinet in accordance with DL-SM-B-682825 shall be installed in the shelter to operate in conjunction with the space heater (see 3.6.2.1). A rear limit stop shall be installed and the overall length of the top film drying rack shall be reduced sufficiently to prevent contact with the wiring bundle. The film drying compartment shall dry 35mm, 120/220, 70mm, and 4 by 5 inch (102 by 127mm) wet, processed film. The print drying compartments shall dry 8 by 10 inch (203 by 254mm) prints at the rate of 100 prints per cycle.

3.6.6.5 Processing trays PH-158( ). Three each processing trays PH-158, conforming to RR-T-646, type I, size 1, shall be furnished.

3.6.6.6 Processing tray PH-161( ). One processing tray PH-161 conforming to RR-T-646, type I, size 2, shall be furnished.

3.6.7 Captioning equipment. The captioning equipment, consisting of a typewriter, shall be installed in accordance with DL-SM-B-682858.

3.6.8 Timer FN-5( ). Two timers FN-5( ) in accordance with DL-SM-B-682911 shall be furnished.

3.6.9 Timer FM-17( ). One timer FM-17( ) in accordance with DL-SM-B-682912 shall be furnished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.6.10 Refrigerator and cabinet.. A refrigerator and cabinet shall be installed in accordance with DL-SM-B-682856. The hole size in the corner of the counter top shall be increased sufficiently to allow passage of the power cable plug of the refrigerator. The refrigerator shall maintain an internal temperature of 45°F ±5°F (7°C ±3°C) with an external ambient temperature of 100°F (38°C).

3.7 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be in accordance with requirement 7 of MIL-STD-454.

3.8 Operation.

3.8.1 Functional operation. The following darkroom components shall operate within the limits stated below:

a. Switches on the following components shall start and stop the operation of the equipment:
Air conditioner.
Dehumidifier.
Exhaust fans.
Heater-dryer.
Lighting fixtures.
Intercommunication station LS-586/U.
Printer, projection, photographic EN-105( ).
Refrigerator.
Safelight, darkroom, photographic FM-188( ).
Processing machine, photographic paper EH-91( ).
Presoaker, photographic film FM-187( ).

b. The following timers shall be accurate within 20 percent when operated for 30 seconds, and for one minute periods:
Timer, interval FM-17( ).
Timer, interval FN-5( ).
Timer, stop FM-103( ).

c. The water system and the following components shall not leak and shall operate using unpressurized water:
Water supply pump.
Storage tank.
Hot water heater.
Temperature control unit, photographic processing FH-19( ).
Water circulating pump.

3.8.2 Performance. The darkroom shall perform the following photographic functions.

3.8.2.1 Developing.

a. Processing 35mm and 120/220 color reversal roll film using processing unit, photographic film EJ-13( ).
b. Conventional processing of 70mm black and white roll film using processing unit, photographic film EJ-12( ).

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

c. Conventional processing of 4 by 5 inch (102 by 127mm) black and white film packs and cut film using 3 developing tanks, each 1 quart (0.95 liter) capacity, measuring 2.5 inch (63.5mm) by 5.375 inch (136.5mm) by 5.813 inch (147.6mm) deep, Leedal, Inc. P/N 321, or equal.
d. Conventional processing of 35mm and 120/220 black and white roll film using processing unit, photographic film EJ-13( ).
e. Conventional processing of 8 by 10 inch (203 by 254mm) enlargement prints using PH-158( ) photographic trays.
f. Conventional processing of 11 by 14 inch (279 by 356mm) enlargement prints using PH-161( ) photographic trays.
g. Processing 8 by 10 inch (203 by 254mm) enlargement prints using processing machine, photographic paper EH-91( ).

3.8.2.2 Printing. Printing 35mm, 70mm, 120/220, and 4 by 5 inch (102 by 127mm) black and white negatives by contact or projection, using printer, projection photographic EN-105( ), on stabilization or conventional print paper.

3.8.2.3 Drying.

a. Drying 35mm, 70mm, 120/220, and 4 by 5 inch (102 by 127mm) wet, processed film using the heater-drayer.
b. Drying 8 by 10 inch (203 by 254mm) prints using the heater-dryer.

3.8.2.4 Captioning. Captioning of prints using typewriter and pressure-sensitive paper.

3.9 Environmental.

3.9.1 Temperature.

3.9.1.1 Low. The darkroom shall exhibit no signs of damage, leakage, or operational degradation at a low operating temperature of -40°F (+0 or -6)°F (-40°C (+0 or -3)°C) and a low storage temperature of -60°F (+0° or -6°)°F (-51°C (+0 or -3)°C). While at low operating temperature the space heater shall maintain the darkroom internal temperature as specified in 3.6.2.1.

3.9.1.2 High. The darkroom shall exhibit no signs of damage, leakage, or operational degradation at a high operating temperature of 120°F (+6 or -0)°F (49°C (+3 or -0)°C). While at a high operating temperature the air conditioners shall maintain the darkroom internal temperature as specified in 3.6.2.2 and the dehumidifier shall maintain the darkroom humidity as specified in 3.6.2.3.

3.9.2 Relative humidity. When exposed to a relative humidity of 90 percent the darkroom shall exhibit no physical damage such as corrosion, rust blistering or swelling, no deterioration of parts and material, or operational degradation.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

3.9.3 Altitude (elevation). When tested as specified in 4.6.6.4, the darkroom shall exhibit no degradation of functional operations specified herein.

3.9.4 Rain. When tested as specified in 4.6.6.5, there shall be no evidence of leakage in the shelter.

3.9.5 Fording. When tested as specified in 4.6.6.6, the structure shall show no leakage or damage.

3.9.6 Drop. When tested as specified in 4.6.6.7.1 and 4.6.6.7.2, there shall be no evidence of permanent damage to the contents of the shelter.

3.9.7 Bounce. When tested as specified in 4.6.6.8 there shall be no sustaining damage to the darkroom.

3.9.8 Shock, railroad humping. When tested as specified in 4.6.6.9 there shall be no sustaining damage to the darkroom.

3.9.9 Fungus. When tested as specified in 4.6.6.10 the darkroom and installed equipment shall show no more than six minute unrelated spots, each no greater than 0.015 in^2 (9.7mm^2) in area, of sparse microbial growth as evidenced by growth colonization (which includes branching and sporulation on or within each cubic foot (0.0028m^3), or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination shall not be included in this evaluation (see 6.4).

3.10 Electromagnetic interference characteristics. The darkroom shall conform to the electromagnetic interference requirements specified in MIL-STD-461, class IIB.

3.11 Light tightness. When tested as specified in 4.6.8, there shall be no difference in the density of the two groups of film.

3.12 Finish. The shelter and components shall be finished in accordance with the drawings and MIL-F-14072.

3.13 Marking. Marking shall be in accordance with MIL-M-13231.

3.13.1 Visibility. Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with the minimum disassembly of equipment.

3.13.2 Serial numbers. The following components shall be serial numbered:

Darkroom, photographic, transportable ES-82( ).
Temperature control unit, photographic processing FH-19( ).
Processing machine, photographic paper EH-91( ).
Processing unit, photographic film EJ-13( ).
Safelight, darkroom, photographic FM-188( ).
Timer, interval FN-5( ).
Heater, water.
Heater, space, multifuel, with blower, 60,00 Btu/hr.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currentversion before use.

Printer, projection, photographic EN-105( ).
Processing unit, photographic film EJ-12( ).
Intercommunication station LS-586/U.
Timer, interval FM-17( ).

3.13.3 Wiring diagram. Circuit labels in accordance with SM-C-682477 shall be fastened to the inside and the outside of the cover of the circuit breaker box. A label in accordance with SM-D-682338 shall also be fastened to the outside of the cover.

3.13.4 Plumbing diagram. A plumbing diagram in accordance with SM-D-682481 shall be installed in the front compartment.

3.13.5 Safety label. A caution label in accordance with SM-C-682850 shall be fastened on the refregerator door.

3.13.6 Safety stenciling. The following stencils, in capital letters, shall be applied as follows:

a. DO NOT OPERATE HEATER OR DRYER THERMOSTAT WITHOUT OPENING DAMPER. This notice shall be stenciled adjacent to the drying cabinet in yellow paint.

b. NO SMOKING. This notice shall be stenciled above each internal door in red paint.

c. PRESS TO OPEN. This notice shall be stenciled adjacent to the circuit breaker box in black paint.

d. WARNING - DANGEROUS VOLTAGES EXIST IN THIS EQUIPMENT. This notice shall be stenciled, in a conspicuous place in each compartment, in red paint.

3.13.7 External marking. Stencils in accordance with SM-E-682593 shall be marked on the outside of the shelter.

3.13.8 Aircraft loading data plate. The aircraft loading data plate on the bare shelter shall be replaced with an aircraft loading data plate in accordance with SC-D-82768.

3.13.9 Additional marking.

3.13.9.1 Heater access panel. The heater access panel shall be marked to indicate that it is an access to the space heater and the fresh air sliding intake.

3.13.9.2 Drain valves. The drain valves D5, D10, and D11 shall be labeled and color coded.

3.13.9.3 Compartment light switches. The compartment light switches shall be labeled.

3.13.9.4 Electrical outlet boxes. Each outlet box shall be labeled to identify the plug-in unit assigned.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currentversion before use.

3.13.9.5 Exhaust fan fuseholder. The fuseholder for the front compartment exhaust fan shall be labeled.

3.13.9.6 Air conditioner thermostats. Both thermostats for the air conditioners shall be labeled.

3.13.9.7 Heater fresh-air sliding door. The heater fresh-air sliding door shall be labeled.

3.13.9.8 Heater louver control. The heater louver control shall be labeled for identification, function and location.

3.13.9.9 Escape door. The inside and outside of the escape door shall be marked (a) to indicate the direction of handle rotation to open the door; (b) to identify the door as an emergency exit; and (c) to warn that the door shall not be blocked.

3.13.9.10 Sink cover stowage hanger. The sink cover stowage hanger shall be identified by means of a stencil.

3.13.9.11 Top storage drawer screw latches. An arrowline indicator shall be marked around each top storage drawer screw latch to indiate the direction of knob rotation for locking or unlocking the drawer.

3.14 Government furnished property. When specified (see 6.2c) the Government shall furnish the following (see 6.5):

a. Shelter, electrical equipment S-280( )/G, MIL-S-55286, 1 required (see 3.6.1).
b. Heater, space, MIL-H-11511, 1 required (see 3.6.2.1).
c. Air conditioner, MIL-A-14372, 2 required (see 3.6.2.2).
d. Flashlight, MX-991/U, 1 required.
e. Ground rod, MX-148/U, 1 required.

3.15 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454, and the applicable portions of the following:

a. In MIL-P-11268:
Cable, portable power, color coding.
Screws, other threaded devices, and related parts.
Brazing.
Cleaning.
Riveting.
Soldering.
Treatment, overall, (moisture and fungus proofing treatment (MFP) on equipment of established design).
Wiring and cabling, including:
Cabling.
Slack.
Protection.
Support.
Clearance.
Splicing and stretching.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currentversion before use.

b. In this specification:

3.4.1	Threaded inserts.
3.5.1	Physical construction.
3.5.2	Installation.
3.6.1.1	Escape door.
3.6.1.2	Floor covering.
3.6.1.3	Partitions.
3.6.1.4	Shelves.
3.6.1.5	Ladder.
3.6.2.2	Air conditioner covers.
3.6.2.3	Dehumidifier, space.
3.6.3	Electrical system.
3.6.3.1	Identification of wiring.
3.6.3.3	Power cables.
3.6.3.4	Lighting fixtures.
3.6.4	Plumbing system.
3.6.4.1	Piping.
3.6.4.2	Water supply pump.
3.6.4.5	Hot water heater.
3.6.4.6	Temperature control unit, photographic FH-19( ).
3.6.4.6.1	Valve control.
3.6.4.7	Sink.
3.6.6.3	Dust covers.
3.6.6.4	Heater-dryer.
3.6.10	Refrigerator and cabinet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspections (see 4.5).
d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article unit. When required and as specified in the contract, the contractor shall furnish first article units of the complete darkroom.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for group A, group B, and group C (see tables I, II, IIII respectively). The inspection shall be performed in the following order: 4.4, group A and group B for all units, and group C as specified in table III. After completion of group C environmental tests, conforming units shall be reinspected and shall pass all group A inspections. Failure of any inspection shall be cause for rejection of the first article.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requirements.

Item	Where required
Shelter, elect. equip. S-280( )/G	3.6.1
Heater, space	3.6.2.1
Air conditioners	3.6.2.2
Electrical system	3.6.3
Safelight FM-188( )	3.6.3.5
Printer, projection, photo EN-105( )	3.6.6.1
Processing tray PH-158( )	3.6.6.5
Processing tray PH-161( )	3.6.6.6
Finish	3.12
Marking	3.13 thru 3.13.9.11

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsiblity for performing any additional inspection which is necessaary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in table I. Discrete lots shall be formed from units that pass the inspection. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and acceptable quality levels (AQL) indicated in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspections

Inspection	Requirement paragraph	Test paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Visual and mechanical:				
Darkroom, photo, transportable ES-82( )	3.5.1, 3.15	4.6.9	4.0	10.0
Air ducts	3.5.1, 3.15	4.6.9	1.0	4.0
Intercom LS-586/U	3.5.1, 3.15	4.6.9	2.5	6.5
Heater-dryer	3.6.2.1, 3.6.6.4, 3.15	4.6.9	1.0	4.0
Dehumidifier	3.6.2.3, 3.15	4.6.9	2.5	6.5
Electrical system	3.6.3, 3.15	4.6.9	1	4.0
Power cables	3.6.3.3, 3.15	4.6.9	1.0	4.0
Lighting fixtures	3.6.3.4, 3.15	4.6.9	1.0	4.0
Storage tank	3.6.4.4, 3.15	4.6.9	1.0	4.0
Water heater	3.6.4.5, 3.15	4.6.9	1.0	4.0
Temperature control unit FH-19( )	3.6.4.6, 3.15	4.6.9	2.5	6.5
Sink	3.6.4.7, 3.15	4.6.9	1.0	4.0
Processing unit EJ-12( )	3.6.5.1, 3.15	4.6.9	2.5	6.5
Processing unit EJ-13( )	3.6.5.2, 3.15	4.6.9	2.5	6.5
Processing EH-91( )	3.6.6.2, 3.15	4.6.9	2.5	6.5
Dust covers	3.6.6.3, 3.15	4.6.9	2.5	6.5
Timer FN-5( )	3.6.8, 3.15	4.6.9	2.5	6.5
Timer FM-17( )	3.6.9, 3.15	4.6.9	2.5	6.5
Cabinet	3.6.10, 3.15	4.6.9	1.0	4.0
Installation of:				
Threaded inserts	3.4.1	4.6.1.1	1.0	1/
Storage box	3.5.2, 3.15	4.6.9	1.0	4.0
Escape door	3.6.1.1, 3.15	4.6.9	1.0	4.0
Floor covering	3.6.1.2, 3.15	4.6.9	1.0	4.0
Partitions	3.6.1.3, 3.15	4.6.9	1.0	4.0
Counters and shelves	3.6.1.4, 3.15	4.6.9	1.0	4.0
Air conditioners	3.6.2.2, 3.15	4.6.9	1.0	4.0
Plumbing	3.6.4, 3.15	4.6.9	1.0	4.0
EN-105( ) and EH-91( )	3.6.6.1, 3.6.6.2, 3.15	4.6.9	1.0	4.0
Typewriter	3.6.7, 3.15	4.6.9	1.0	4.0
Refrigerator and cabinet	3.6.10, 3.15	4.6.9	1.0	4.0
Operational, functional	3.8.1	4.6.5.1	1.0	1/
Light tightness	3.11	4.6.8	1.0	1/

1/ All defects shall be considered major.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the Government, except that the operational tests in 4.6.5.1 shall be last.

4.5.2 Group B inspection. Group B inspection shall be as specified in table II. Group B inspections shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL (percent defective)
Plumbing system	3.6.4	4.6.3	4.0 for the group
Piping	3.6.4.1	4.6.3.1	
Pump	3.6.4.2	4.6.3.2	
Pressure regulator	3.6.4.3	4.6.3.3	
Liquid level switch	3.6.4.5.1	4.6.3.3	
Pressure relief valve	3.6.4.5.2	4.6.3.3	
Drainage	3.6.4.8	4.6.3.5	
Temperature control unit FH-19( )	3.6.4.6	4.6.3.4	6.5
Interchangeability	3.7	4.6.4	6.5

4.5.2.1 Group B sampling plans. Sampling for group B inspection shall be in accordance with MIL-STD-105, inspection level S-4. AQL shall be as specified in table II.

4.5.2.2 Order of inspection within Group B. Group B inspection shal be performed in an order satisfactory to the government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table III. Samples shall be selected in accordance with 4.5.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

TABLE III. Group C inspection.

Inspection	Requirement paragraph	Inspection paragraph
Subgroup 1		
Electromagnetic interference	3.10	4.6.7
Performance	3.8	4.6.5.2
Threaded inserts	3.4.1	4.6.1.2
Temperature	3.9.1	4.6.6.1 - 4.6.6.2.1
Elevation	3.9.3	4.6.6.4
Humidity	3.9.2	4.6.6.3
Shock, drop	3.9.6	4.6.6.7
Performance	3.8	4.6.5.2
Subgroup 2		
Performance	3.8	4.6.5.2
Shock, railroad humping	3.9.8	4.6.6.9
Bounce	3.9.7	4.6.6.8
Rain	3.9.4	4.6.6.5
Fording	3.9.5	4.6.6.6
Fungus 1/	3.9.9	4.6.6.10
Performance	3.8	4.6.5.2

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.5.3.1 Group C sampling plan. One sample darkroom from every 50 or fraction thereof produced shall be selected at random for each subgroup of inspections in table III. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in the order shown in each subgroup in table III.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract (see 6.2d).

4.5.4 Reinspection of conforming group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both group B and group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Methods of inspection.

4.6.1 Threaded insert.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currentversion before use.

4.6.1.1 Production test. Install an eyebolt in the insert to be tested. Attach a turnbuckle and a dynamometer to the eyebolt and anchor the dynamometer to a similar insert or other suitable device in the opposite panel of the shelter. Tighten the turnbuckle to obtain a 1,000 pound (454kg) tensile load on the test insert. After the load has been removed, install a bolt in the threaded insert and apply a torque wrench to the bolt to obtain a load of 40 lbf. in. (4.5 Nm).

4.6.1.2 Verification test. Same as above except 2,000 pound (907kg) load shall be applied.

4.6.1.3 Failure criteria. Nonconformance to 3.4.1 shall constitute failure of this test.

4.6.2 Electrical tests.

4.6.2.1 Breakdown voltage test. A voltage of twice the rated voltage plus 1,000 volts shall be applied for one minute between all shelter wiring and adjacent conductors not in the circuit and between all shelter wiring and ground. The test potential may be direct current or peak alternating current of the same value. Nonconformance to 3.6.3.2 shall constitute failure of this test.

4.6.2.2 Insulation resistance. The assembled components of the power cable assembly (exclusive of the shelter and power unit ends) shall be immersed in water for 24 hours and the breakdown voltage test (4.6.2.1) shall be performed between phases and ground. Nonconformance to 3.6.3.3.1 shall constitute failure of this test.

4.6.3 Plumbing system.

4.6.3.1 Piping. All outlets in the piping system shall be closed and the system subjected to 90 psi (621kPa) pressure. The filter shall be removed or by-passed during this test. Nonconformance to 3.6.4.1 shall constitute failure of this test.

4.6.3.2 Water supply pump. Using a water source not less than 15 feet (4.57m) below the level of the pump, operate the pump at a 100 foot (30m) head pressure. Pump water through the entire plumbing system. Observe the automatic pressure-activated switch at the high and low level limits of the water storage tank. Nonconformance to 3.6.4.2 shall constitute failure of this test.

4.6.3.3 Controls. The following controls shall be tested to determine compliance with the specified requirements:

Control	Requirement paragraph
Pressure regulator	3.6.4.3
Hot water thermostate	3.6.4.5, 3.6.4.6.2
Liquid level switch	3.6.4.5.1
Pressure relief valve	3.6.4.5.2

Inability of the controls to meet the requirements as specified above shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.6.3.4 Temperature control unit, photographic processing FH-19( ). Place the temperature-controlled compartment of the sink in an ambient temperature of 100°F (38°C) and add 26 gallons (98 liters) of water. Operate the pump at a 4 psi (28kPa) head pressure and heat the water to 90°F (32°C). During the test the water circulation rate shall be monitored and the pre-set water temperature shall be compared with the actual temperature of the circulating water. Nonconformance to 3.6.4.6, 3.6.4.6.1, 3.6.4.6.2, or 3.6.4.6.3 shall constitute failure of this test.

4.6.3.5 Drainage. With the shelter in a level position and filled with water or chemical solutions, attach a 50 foot (15m), 1 inch (25.4mm) inside diameter hose to the drain. Position the discharge end of the hose not more than 2 feet (0.61m) lower than the floor of the shelter and drain by gravity all water or chemical solutions. Nonconformance to 3.6.4.8 shall constitute failure of this test.

4.6.4 Dimensional interchangeability. The dimensions listed below shall be measured to determine compliance with the physical interchangeability requirements of 3.7. When a dimension is not within specified limits, it shall be considered a major defect.

a. Location of threaded inserts.
b. Location of hinges.
c. Location of fasteners.

4.6.5 Operational tests.

4.6.5.1 Functional operation. The following components of the darkroom shall be tested as follows to demonstrate operational capability:

a. Switches shall be turned ON and OFF on the following equipment:

Air conditioner.
Dehumidifier.
Exhaust fans.
Heater-dryer.
Lighting fixtures.
Intercommunication station LS-586/U.
Printer, projection, photographic EN-105( ).
Refrigerator.
Safelights, darkroom, photographic FM-188( ).
Processing machine, photographic paper EH-91( ).

Nonconformance of any component to 3.8.1a shall constitute failure of this test.

b. The following timers shall be set for 30 seconds and one minute intervals and checked with a stop watch for operation:

Timer, interval FM-17( ).
Timer, interval FN-5( ).
Timer, stop FM-103( ).

Nonconformance of any timer to 3.8.1b shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

c. The system shall be filled with unpressurized water to demonstrate the operational capability of the following equipment:

Water supply pump.
Storage tank.
Hot water heater.
Temperature control unit, photographic processing FH-19( ).
Water circulating pump.

The system shall be filled with water and the equipment switched ON and OFF. Two 55-gallon (208 liters) drums filled with water and two empty 55-gallon (208 liters) drums may be used to accommodate the water source and drain. Nonconformance of any equipment to 3.8.1c shall constitute failure of this test.

4.6.5.2 Performance test. The darkroom shal be tested as follows:

a. One exposed roll of 35mm of 120 color film shall be completely processed using the E-4 color process.

b. One exposed roll of 70mm film and one 4 by 5 inch (102 by 127mm) cut film shall be completely processed using conventional black and white processing. The processed film shall be projection printed on each of the photographic EN-105( ) printers, (one 8 by 10 inch) (203 by 254mm) print from the roll and one from the cut film) using conventional wet process and including the use of the film and print drying cabinet.

c. Two 8 by 10 inch (203 by 254mm) prints shall be processed through each photographic paper processing machine EH-91( ).

d. Captioning equipment shall be operated using the typewriter and pressure-sensitive paper.

Nonconformance to 3.8.2 shall constitute failure of this test.

4.6.6 Environmental.

4.6.6.1 Low temperature. The darkroom prepared for operation with its support equipment, shall be tested in accordance with method 502.1, procedure I, of MIL-STD-810. The storage temperature shall be -60°F (+0 or -6)°F (-51°C (+0 or -3)°C) and the operating temperature shall be -40°F (+0 or -6)°F (-40°C (+0 or -3°C). The doors shall remain open except during the frigid operation test. Nonconformance to 3.9.1.1 shall constitute failure of this test.

4.6.6.1.1 Frigid operation test. With the ambient exterior temperature at -40°F (-40°C), the heater shall be energized and the film and print dryer shall be set for maximum operation. The interior temperature shall be measured by means of thermocouples, located 12 inches (305mm) form the ceiling and 12 inches (305mm) from the floor, in each compartment. Nonconformance to 3.9.1.1 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currentversion before use.

4.6.6.2 High temperature. The darkroom prepared for operation with its support equipment, shall be tested in accordance with method 501.1, procedure II, of MIL-STD-810. The operating temperature shall be 120°F (+6 or -0)°F (49°C (+3 or -0)°C). The doors shall remain open except during the tropic operation test. Nonconformance to 3.9.1.2 shall constitute failure of this test.

4.6.6.2.1 Tropic operation test. With the ambient exterior temperature at 120°F (49°C), the air conditioners and the dehumidifier shall be energized. The interior temperature shall be measured by means of thermocouples, located 12 inches (305mm) from the ceiling and 12 inches (305mm) from the floor, in each compartment. With the ambient exterior temperature at 100°F (38°C) and the relative humidity increased to 90 percent, the relative humidity of the air inside the darkroom shall be measured. Nonconformance to 3.9.1.2 shall constitute failure of this test.

4.6.6.3 Humidity tests. The darkroom shall be subjected to the humidity test specified in method 507.1, procedure II, of MIL-STD-810, with test measurements during Step 4 and the last 5-hour period of the fifth 48-hour cycle in Step 6. The doors of the shelter shall remain open during cycling. Nonconformance to 3.9.2 shall constitute failure of this test.

4.6.6.4. Altitude (elevation) test. The darkroom shall be tested in accordance with method 500.1, procedure I, of MIL-STD-810. The doors of the shelter shall remain open except when the operation test is performed. Nonconformance to 3.9.3 shall constitute failure of this test.

4.6.6.5 Rain test. The darkroom shall be subjected to part (a) of the watertightness test specified in MIL-S-55286, except that the bottom row of nozzles shall be eliminated with the exception of the nozzles facing the door end panel. The operation of the latching mechanism shall not be required. Nonconformance to 3.9.4 shall constitute failure of this test.

4.6.6.6 Fording. The darkroom, with all openings closed, shall be immersed in water to a depth of 21 inches (533mm), measured from the bottom of the shelter skids, for a period of 2 hours. The darkroom shall be opened after the immersion period and inspected. Nonconformance to 3.9.5 shall constitute failure of this test.

4.6.6.7 Drop tests.

4.6.6.7.1 Flat drop. The darkroom shall be lifted 18 inches (457mm) from the ground and allowed to drop freely with the skids impacting onto a concrete surface. This test shall be performed once. Nonconformance to 3.9.6 shall constitute failure of this test.

4.6.6.7.2 Rotational drop. The impact test shall be performed on each bottom edge of the shelter. The darkroom shall have one edge placed on a nominal 4 inch (102mm) timber, with the timber placed under the skids. The opposite edge of the shelter shall be raised 18 inches (457mm), measured from the bottom of the skid, and allowed to drop freely onto a concrete surface. Nonconformance to 3.9.6 shall constitute failure of this test.

4.6.6.8 Bounce test. The darkroom shall be tested in accordance with method 514.2, procedure XIII of MIL-STD-810. Nonconformance to 3.9.7 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

4.6.6.9 Shock test, railroad humping. The darkroom shall be loaded on a flatcar in the manner normally used for transportation without an exterior protective crate. The contents shall include the items covered by this test. No bracing, dunnage, blocking, or other protective means shall be used inside the shelter, unless a permanent part of the equipment. The shelter shall be secured to the flatcar, as specified in MIL-S-55286. The shelter shall then be tested in accordance with method 516.2, procedure VI of MIL-STD-810. The order in which the impacts are performed shall be selected by the contracting officer. The impact shall be repeated if the blocking which holds the shelter in place is torn loose. Nonconformance to 3.9.8 shall constitute failure of this test.

4.6.6.10 Fungus. The darkroom shall be subjected to the fungus test in accordance with method 508.2, of MIL-STD-810. There shall be abundant growth colonization (6.4.2) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the darkroom is permitted for 72 hours prior to the fungus test. Nonconformance to 3.9.9 shall constitute failure of this test.

4.6.7 Electromagnetic interference characteristics. The darkroom shall be tested in accordance with MIL-STD-462 for electromagnetic interference. The test shall be conducted with the doors closed. Nonconformance to 3.10 shall constitute failure of this test.

4.6.8 Light leak test. The darkroom shall be tested for light leaks as follows:

a. The darkroom shall be prepared for operation by uncovering all required openings. The darkroom shall be subjected to an intensity of light approximating direct sunlight at noontime.
b. Photographic film having an ASA rating of 100 shall be divided into two groups. One group shall be placed, uncovered, on each work area in each compartment for a period of 30 minutes. The other group shall be placed in total darkness.
c. After the 30 minute period, both groups of film shall be developed simultaneously using normal processing procedures, and then examined for any difference in density, using a standard laboratory densitometer. Any fogging due to light leak shall constitute failure of this test.

4.6.9 Visual and mechanical. The darkroom, including all accessories shall be examined for the defects listed in MIL-STD-252 and table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of visual and mechanical defects.

Classification	Defects
Major	1. Tie down straps missing. 2. Sharp projections on equipment. 3. Nonconformance of labels and marking of equipment. 4. Punctures in interior skin of shelter. 5. Missing or inoperative component. 6. Foreign material which affect operation. 7. Taut wiring with possiblity of breaking. 8. Compartment doors sag or stick against frame, or bottom edges strike floor.
Minor	1. Cracked metal not affecting use. 2. Taut wiring with no possiblity of breaking. 3. Useless hoses or slots. 4. Foreign material in equipment not affecting operation. 5. Scratches or abrasions on finish.

4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table V, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

TABLE V. Packing inspection provisions.

No.	Characteristic	Method ofinspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-S-55507.

6. NOTES

6.1 Intended use. The darkroom is used as a complete self-contained phtotographic laboratory for processing film and paper.

6.1.1 Power supply. Electrical power to the darkroom shall be supplied by utilizing generator set, diesel engine, trailer mounted PU-405/M or a 120/208 VAC ±5 percent, 60 Hz ±5 percent, 3 phase 4 wire power source.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2).
c. Action required relative to group C failures (see 4.5.3.3).
d. Level A or B preservation and packing (see section 5).
e. When first article inspection rough handling tests are not required.
f. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.
g. Government furnished property required (see 3.14).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Definitions.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currentversion before use.

SHOCK, railroad humping 3.9.2

6.4.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.4.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.4.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.4.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.4.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5 Government-furnished property. The contracting officer should arrange to furnish the following:

Shelter, Electrical Equipment S-280( )/G (See 3.6.1)
Heater, Space (See 3.6.2.1)
Air Conditioners (See 3.6.2.2)
Flashlight MX-991/U
Ground Rod MX-148/U

6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example ES-82W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.7 Level of preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.8 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ER

Preparing activity:
Army -ER
Project 6780-A014

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currentversion before use.

6.8.1.2 shall constitute failure of this test.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Headquarters
US Army Comabt Surveillance and
Target Acquisition Laboratory
ATTN: DELCS-PE
Fort Monmouth, New Jersey 07703

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T03:58Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 PREVIOUS EDITIONS ARE OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 20240-26103:582
Check the source to verify that this is the current version before use.